RUNBOOK — DevOps Assessment (Maachao-Assignment)

Owner / Environment

Owner: Amit Shah

Domain: testingproject.online

SSH Port: 2222

Public Ports: 80/443 only for Nginx (plus Grafana if enabled on 3001)

Task 1 — Server Hardening

Create deploy user and move SSH access Commands:

```
sudo adduser deploy
sudo usermod -aG sudo deploy (Add user deploy to the sudo group)
```

Copy SSH keys (ubuntu -> deploy)

```
sudo mkdir -p /home/deploy/.ssh
sudo cp /home/ubuntu/.ssh/authorized_keys /home/deploy/.ssh/authorized_keys
sudo chown -R deploy:deploy /home/deploy/.ssh
sudo chmod 700 /home/deploy/.ssh
sudo chmod 600 /home/deploy/.ssh/authorized_keys
```

sudo vi /etc/ssh/sshd_config **(SSH hardening)**

- Port 2222
- PasswordAuthentication no
- PermitRootLogin no
- PubkeyAuthentication yes

```
Last login: Fri Mar  6 03:54:31 2026 from 165.85.130.48
deploy@ip-172-31-39-3:~$ cat vi /etc/ssh/sshd_config
cat: vi: No such file or directory

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
```

Validate SSH config and restart ssh

sudo sshd –t

sudo systemctl restart ssh

UFW firewall (allow only required)

sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 2222/tcp
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw allow 51820/udp
sudo ufw enable
sudo ufw status verbose

sudo systemctl restart ssh

Verify port

sudo ss -tulpn | grep ssh (but still port 22 is showing)

```
deploy@ip-172-31-39-3:~$ sudo ss -tulpn | grep ssh
tcp   LISTEN 0      4096           0.0.0.0:22        0.0.0.0:*    users:(("sshd",pid=16609,fd=3),("systemd",pid=1,fd=105))
tcp   LISTEN 0      128          127.0.0.1:6010      0.0.0.0:*    users:(("sshd",pid=15289,fd=8))
tcp   LISTEN 0      4096              [::]:22           [::]:*    users:(("sshd",pid=16609,fd=4),("systemd",pid=1,fd=106))
tcp   LISTEN 0      128             [::1]:6010         [::]:*    users:(("sshd",pid=15289,fd=7))
deploy@ip-172-31-39-3:~$
```

Reloaded the daemon and restarted the ssh.socket .

sudo systemctl daemon-reload
sudo systemctl restart ssh.socket
sudo systemctl restart ssh

```
deploy@ip-172-31-39-3:~$ sudo ss -tulpn | grep ssh
[sudo] password for deploy:
tcp   LISTEN 0      4096           0.0.0.0:2222      0.0.0.0:*    users:(("sshd",pid=18570,fd=3),("systemd",pid=1,fd=68))
tcp   LISTEN 0      128          127.0.0.1:6010      0.0.0.0:*    users:(("sshd",pid=19824,fd=8))
tcp   LISTEN 0      4096              [::]:2222         [::]:*    users:(("sshd",pid=18570,fd=4),("systemd",pid=1,fd=69))
tcp   LISTEN 0      128             [::1]:6010         [::]:*    users:(("sshd",pid=19824,fd=7))
deploy@ip-172-31-39-3:~$
```

```
AzureAD+AmitShah@DESKTOP-GOIMMO8 MINGW64 ~/Downloads
$ ssh -i "test-server.pem" -p 22 deploy@13.234.239.76
ssh: connect to host 13.234.239.76 port 22: Connection timed out

AzureAD+AmitShah@DESKTOP-GOIMMO8 MINGW64 ~/Downloads
$ ssh -i "test-server.pem" -p 2222 deploy@13.234.239.76
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Mar  6 04:03:14 UTC 2026

  System load:  0.0                Temperature:           -273.1 C
  Usage of /:   12.3% of 18.33GB   Processes:             121
  Memory usage: 33%                Users logged in:       1
  Swap usage:   0%                 IPv4 address for ens5: 172.31.39.3


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


*** System restart required ***
Last login: Fri Mar  6 03:55:11 2026 from 165.85.130.48
deploy@ip-172-31-39-3:~$
```

Fail2Ban

sudo apt install fail2ban -y

check service

sudo systemctl status fail2ban
sudo vi /etc/fail2ban/jail.d/sshd.conf

Add below parameters

[sshd]
enabled = true
port = 2222
backend = systemd
maxretry = 5
findtime = 300
bantime = 3600

```
deploy@ip-172-31-39-3:~$ cat /etc/fail2ban/jail.d/sshd.conf
[sshd]
enabled = true
port = 2222
backend = systemd
maxretry = 5
findtime = 300
bantime = 3600
```

Tets and restart the configuration

sudo fail2ban-client -t
sudo systemctl restart fail2ban
sudo systemctl status fail2ban
sudo fail2ban-client status sshd

Test ban:

- attempt wrong ssh login 5 times

Verify:

- sudo fail2ban-client status sshd

Unban:

- sudo fail2ban-client set sshd unbanip 13.234.239.76

```
deploy@ip-172-31-39-3:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
deploy@ip-172-31-39-3:~$
```

Automatic security updates (security only no auto reboot)

sudo apt install unattended-upgrades -y

sudo dpkg-reconfigure unattended-upgrades -> yes

Configure security patches ONLY -> sudo vi /etc/apt/apt.conf.d/50unattended-upgrades

updated the setting like:-

Unattended-Upgrade::Allowed-Origins {
"${distro_id}:${distro_codename}-security";
};

Now Configure Reboot Behavior -> Unattended-Upgrade::Automatic-Reboot "false";

For dry run test
sudo unattended-upgrades --dry-run --debug

to see alert reboot is required or not
cat /var/run/reboot-required

Task 2 — DOCKER + APPLICATION STACK

Install docker and docker compose on the server
Using below commands

```
sudo apt update
sudo apt install -y ca-certificates curl gnupg

sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg

echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-
compose-plugin

sudo usermod -aG docker deploy
newgrp docker

docker --version
docker compose version
```

Dockerfile

```
FROM python:3.12-slim AS builder
WORKDIR /build
COPY app/requirements.txt .
RUN pip install --no-cache-dir --prefix=/install -r requirements.txt

FROM python:3.12-slim
WORKDIR /app
COPY --from=builder /install /usr/local
COPY app /app
EXPOSE 8000
HEALTHCHECK --interval=10s --timeout=3s --retries=3 \
  CMD python -c "import urllib.request; urllib.request.urlopen('http://127.0.0.1:8000/health')"
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

docker-compose.yml

```
services:
  api:
    build: .
    container_name: api
    restart: always
    networks: [backend]
    expose:
      - "8000"

  postgres:
    image: postgres:16
    container_name: postgres
    restart: always
    env_file: .env
    environment:
      POSTGRES_DB: ${POSTGRES_DB}
      POSTGRES_USER: ${POSTGRES_USER}
      POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
    volumes:
      - pgdata:/var/lib/postgresql/data
    networks: [backend]
    expose:
      - "5432"
```

```
  redis:
    image: redis:7-alpine
    container_name: redis
    restart: always
    env_file: .env
    command: ["redis-server", "--requirepass", "${REDIS_PASSWORD}"]
    networks: [backend]
    expose:
      - "6379"

networks:
  backend:

volumes:
  pgdata:
```

Updated the .env file (update your password in postgres password and redis password)

```
POSTGRES_DB=assessment_db
POSTGRES_USER=app_user
POSTGRES_PASSWORD=Updateyourpassword
REDIS_PASSWORD= Updateyourpassword
```

Run the Stack

```
docker compose up -d –build
```

Docker ps

Test health endpoint

curl http://localhost/health

```
deploy@ip-172-31-39-3:~/devops-assessment$ curl http://localhost/health
{"status":"ok"}deploy@ip-172-31-39-3:~/devops-assessment$
deploy@ip-172-31-39-3:~/devops-assessment$
deploy@ip-172-31-39-3:~/devops-assessment$
```

Docker ps output

```
deploy@ip-172-31-39-3:~/devops-assessment$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                    PORTS      NAMES
e39f18a9df8e   postgres:16            "docker-entrypoint.s…"   13 seconds ago   Up 12 seconds             5432/tcp   postgres
457e631e425e   redis:7-alpine         "docker-entrypoint.s…"   13 seconds ago   Up 12 seconds             6379/tcp   redis
3c8b84d02726   devops-assessment-api  "uvicorn main:app --…"   13 seconds ago   Up 12 seconds (healthy)   8000/tcp   api
deploy@ip-172-31-39-3:~/devops-assessment$
```

Auto-restart verification

```
3. test - server    4. test - server
deploy@ip-172-31-39-3:~/devops-assessment$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS                   PORTS      NAMES
0a8c34e4b662   postgres:16            "docker-entrypoint.s…"   7 minutes ago   Up 7 minutes             5432/tcp   postgres
01f79015a54a   redis:7-alpine         "docker-entrypoint.s…"   7 minutes ago   Up 7 minutes             6379/tcp   redis
9e654249160d   devops-assessment-api  "uvicorn main:app --…"   7 minutes ago   Up 7 minutes (healthy)   8000/tcp   api
deploy@ip-172-31-39-3:~/devops-assessment$ docker exec -it api sh
# ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0   1056   732 ?        Ss   06:39   0:00 /sbin/docker-init -- uvicorn main:app --host 0.0.0.0 --port 8000
root           8  0.4  5.0 236068 47736 ?        Sl   06:39   0:01 /usr/local/bin/python3.12 /usr/local/bin/uvicorn main:app --host 0.0.0
root         466  0.9  0.1   2676  1832 pts/0    Ss   06:47   0:00 sh
root         473  0.0  0.4   6788  3968 pts/0    R+   06:47   0:00 ps aux
# kill -9 8
# deploy@ip-172-31-39-3:~/devops-assessment$
deploy@ip-172-31-39-3:~/devops-assessment$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS                            PORTS      NAMES
0a8c34e4b662   postgres:16            "docker-entrypoint.s…"   8 minutes ago   Up 7 minutes                      5432/tcp   postgres
01f79015a54a   redis:7-alpine         "docker-entrypoint.s…"   8 minutes ago   Up 7 minutes                      6379/tcp   redis
9e654249160d   devops-assessment-api  "uvicorn main:app --…"   8 minutes ago   Up 5 seconds (health: starting)   8000/tcp   api
deploy@ip-172-31-39-3:~/devops-assessment$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS                   PORTS      NAMES
0a8c34e4b662   postgres:16            "docker-entrypoint.s…"   8 minutes ago   Up 8 minutes             5432/tcp   postgres
01f79015a54a   redis:7-alpine         "docker-entrypoint.s…"   8 minutes ago   Up 8 minutes             6379/tcp   redis
9e654249160d   devops-assessment-api  "uvicorn main:app --…"   8 minutes ago   Up 17 seconds (healthy)  8000/tcp   api
deploy@ip-172-31-39-3:~/devops-assessment$
```

Task 3 - NGINX REVERSE PROXY + TLS

Reverse proxy

- /api/ -> API container (internal)

- Only Nginx exposed on 80/443

Route53 Hosted Zone for testingproject.online create:
A record: testingproject.online → **EC2 Public IP**

Modified Security Group inbound:
80/tcp from 0.0.0.0/0
443/tcp from 0.0.0.0/0

Confirming only Nginx is public

```
deploy@ip-172-31-39-3:~/devops-assessment$ docker ps
CONTAINER ID   IMAGE                   COMMAND                  CREATED          STATUS                  PORTS
                                                       NAMES
286b1a4b652f   nginx:alpine            "/docker-entrypoint.…"   10 minutes ago   Up 4 minutes            0.0.0.0:80->80/tcp, [::]:80->80/t
cp, 0.0.0.0:443->443/tcp, [::]:443->443/tcp   nginx
0a8c34e4b662   postgres:16             "docker-entrypoint.s…"   3 hours ago      Up 3 hours              5432/tcp
                                                       postgres
01f79015a54a   redis:7-alpine          "docker-entrypoint.s…"   3 hours ago      Up 3 hours              6379/tcp
                                                       redis
9e654249160d   devops-assessment-api   "uvicorn main:app --…"   3 hours ago      Up 3 hours (healthy)    8000/tcp
                                                       api
deploy@ip-172-31-39-3:~/devops-assessment$
```

Nginx config for Reverse Proxy + Redirect + TLS + Headers + Rate limit

Vi nginx/conf.d/default.conf

Update the below config

```
server_tokens off;
limit_req_zone $binary_remote_addr zone=api_ratelimit:10m rate=10r/s;
server {
  listen 80;
  server_name testingproject.online www.testingproject.online;

  location /.well-known/acme-challenge/ { root /var/www/certbot; }

  location / { return 301 https://$host$request_uri; }
}

server {
  listen 443 ssl;
  http2 on;
  server_name testingproject.online www.testingproject.online;
```

```
  ssl_certificate     /etc/letsencrypt/live/testingproject.online/fullchain.pem;
  ssl_certificate_key /etc/letsencrypt/live/testingproject.online/privkey.pem;

  # Security headers
  add_header X-Frame-Options "DENY" always;
  add_header X-Content-Type-Options "nosniff" always;
  add_header Strict-Transport-Security "max-age=31536000; includeSubDomains" always;
  add_header Referrer-Policy "strict-origin" always;

  # make curl -I https://domain show 200
  location = / {
    return 200 "OK\n";
    add_header Content-Type text/plain;
  }

  location /health {
    proxy_pass http://api:8000/health;
  }

  location /api/ {
    # Rate limit
    limit_req zone=api_ratelimit burst=1;
    limit_req_status 429;

    # Proxy to API container (internal only)
    proxy_pass http://api:8000/api/;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
  }
}
```

test the Nginx configuration inside the running container.

```
docker exec -it nginx nginx -t
```

Restarted nginx container

docker compose restart nginx

After restarting the container validated nginx configuration

docker exec -it nginx nginx -t

Run below command to get the TLS certificate

docker compose run --rm certbot certonly \
--webroot -w /var/www/certbot \
-d testingproject.online -d www.testingproject.online \
--email anticamit@gmail.com \
--agree-tos --no-eff-email

Confirm the certificate exist with below command

sudo ls -l nginx/letsencrypt/live/testingproject.online/

```
deploy@ip-172-31-39-3:~/devops-assessment$ sudo ls -l nginx/letsencrypt/live/testingproject.online/
[sudo] password for deploy:
total 4
-rw-r--r-- 1 root root 692 Mar  6 09:35 README
lrwxrwxrwx 1 root root  45 Mar  6 09:35 cert.pem -> ../../archive/testingproject.online/cert1.pem
lrwxrwxrwx 1 root root  46 Mar  6 09:35 chain.pem -> ../../archive/testingproject.online/chain1.pem
lrwxrwxrwx 1 root root  50 Mar  6 09:35 fullchain.pem -> ../../archive/testingproject.online/fullchain1.pem
lrwxrwxrwx 1 root root  48 Mar  6 09:35 privkey.pem -> ../../archive/testingproject.online/privkey1.pem
deploy@ip-172-31-39-3:~/devops-assessment$
```

Verify TLS + Redirect

HTTPS works:

curl -I https://testingproject.online

HTTP redirects to HTTPS:

curl -I http://testingproject.online

Verify Reverse Proxy

curl -sk https://testingproject.online/health

curl -sk https://testingproject.online/api/hello

Use concurrency to trigger rate limit:

seq 1 100 | xargs -P20 -I{} curl -sk -o /dev/null -w "%{http_code}\n"
https://testingproject.online/api/hello | sort | uniq -c

```
deploy@ip-172-31-39-3:~/devops-assessment$ curl -I https://testingproject.online
HTTP/2 200
server: nginx
date: Fri, 06 Mar 2026 09:56:25 GMT
content-type: application/octet-stream
content-length: 3
content-type: text/plain

deploy@ip-172-31-39-3:~/devops-assessment$ curl -I http://testingproject.online
HTTP/1.1 301 Moved Permanently
Server: nginx
Date: Fri, 06 Mar 2026 09:56:31 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive
Location: https://testingproject.online/

deploy@ip-172-31-39-3:~/devops-assessment$ curl -sk https://testingproject.online/health
{"status":"ok"}deploy@ip-172-31-39-3:~/devocurl -sk https://testingproject.online/api/hellonline/api/hello
deploy@ip-172-31-39-3:~/devops-assessment$ for i in $(seq 1 80); do curl -sk -o /dev/null -w "%{http_code}\n" https://testingproject.onli
ne/api/hello; done | sort | uniq -che | sort | uniq -c
     13 200
     67 429
deploy@ip-172-31-39-3:~/devops-assessment$
```

Verify Security Headers

curl -I https://testingproject.online | egrep -i "x-frame-options|x-content-type-options|strict-transport-security|referrer-policy|server"

```
deploy@ip-172-31-39-3:~/devops-assessment$ curl -I https://testingproject.online | egrep -i "x-frame-options|x-content-type-options|stric
t-transport-security|referrer-policy|server"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     3    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
server: nginx
```

TASK 4 — POSTGRESQL ADMIN + BACKUP

logged into PostgreSQL inside the container

docker exec -it postgres psql -U app_user -d assessment_db

\du and \l output

```
deploy@ip-172-31-39-3:~/devops-assessment$ docker exec -it postgres psql -U app_user -d assessment_db
psql (16.13 (Debian 16.13-1.pgdg13+1))
Type "help" for help.

assessment_db=# \du
                                   List of roles
 Role name |                         Attributes
-----------+------------------------------------------------------------
 app_user  | Superuser, Create role, Create DB, Replication, Bypass RLS

assessment_db=# \l
                                                      List of databases
     Name      |  Owner   | Encoding | Locale Provider |  Collate   |   Ctype    | ICU Locale | ICU Rules |  Access privileges
---------------+----------+----------+-----------------+------------+------------+------------+-----------+-----------------------
 assessment_db | app_user | UTF8     | libc            | en_US.utf8 | en_US.utf8 |            |           |
 postgres      | app_user | UTF8     | libc            | en_US.utf8 | en_US.utf8 |            |           | =Tc/app_user        +
               |          |          |                 |            |            |            |           | app_user=CT/app_user
 template0     | app_user | UTF8     | libc            | en_US.utf8 | en_US.utf8 |            |           | =c/app_user         +
               |          |          |                 |            |            |            |           | app_user=CT/app_user
 template1     | app_user | UTF8     | libc            | en_US.utf8 | en_US.utf8 |            |           | =c/app_user         +
               |          |          |                 |            |            |            |           | app_user=CT/app_user
(4 rows)

assessment_db=#
```

ensure app_user can connect only to assessment_db

REVOKE CONNECT ON DATABASE template0 FROM app_user;
REVOKE CONNECT ON DATABASE template1 FROM app_user;
REVOKE CONNECT ON DATABASE postgres FROM app_user;

Check priviledge

SELECT datname, has_database_privilege('app_user', datname, 'CONNECT') AS can_connect
FROM pg_database ORDER BY datname;

pg_hba.conf — allow only docker network

Find pg_hba.conf location

docker exec -it postgres psql -U app_user -d assessment_db -c "SHOW hba_file;"

Find docker network subnet

docker network inspect devops-assessment_backend | grep -A3 Subnet

Note the subnet 172.18.0.0/16

Copy pg_hba.conf into your repo

```
docker cp postgres:/var/lib/postgresql/data/pg_hba.conf ./pg_hba.conf
ls -l ./pg_hba.conf
```

Edit pg_hba.conf to restrict app_user

```
Vi ./pg_hba.conf
```

Add these lines near the top

```
host    assessment_db    app_user    172.18.0.0/16    scram-sha-256
host    assessment_db    app_user    0.0.0.0/0        reject
host    assessment_db    app_user    ::/0             reject
```

Copy back into container and reload config

```
docker cp ./pg_hba.conf postgres:/var/lib/postgresql/data/pg_hba.conf
docker exec -it postgres psql -U app_user -d assessment_db -c "SELECT pg_reload_conf();"
```

Validate if it is working with below command

```
docker exec -it postgres psql -U app_user -d assessment_db -c "SELECT 1;"
```

Automated backup script

Create folders and log file

```
sudo mkdir -p /opt/backup /backups
sudo touch /var/log/backup.log
sudo chmod 755 /opt/backup /backups
sudo chmod 644 /var/log/backup.log
```

Create script /opt/backup/backup-db.sh

```
#!/usr/bin/env bash
set -euo pipefail

TS=$(date +%Y%m%d_%H%M%S)
OUT="/backups/assessment_${TS}.sql.gz"
LOG="/var/log/backup.log"

if docker exec postgres pg_dump -U app_user assessment_db | gzip > "$OUT"; then
echo "$(date -Is) SUCCESS backup created: $OUT" | tee -a "$LOG"
else
echo "$(date -Is) FAIL backup failed" | tee -a "$LOG"
exit 1
fi
```

```
sudo chmod +x /opt/backup/backup-db.sh
```

Add Cron jon daily 2 AM

sudo crontab -e

```
0 2 * * * /opt/backup/backup-db.sh
```

Show cron entry
sudo crontab -l

Manual backup run

```
sudo /opt/backup/backup-db.sh
ls -lh /backups | tail -n 5
tail -n 20 /var/log/backup.log
```

Restore test

Create test table and rows

```
docker exec -it postgres psql -U app_user -d assessment_db -c "CREATE TABLE IF NOT EXISTS test_items(id serial primary key, name text); INSERT INTO test_items(name) VALUES('a'),('b'),('c');"
```

Confirm count:

```
docker exec -it postgres psql -U app_user -d assessment_db -c "SELECT count(*) FROM test_items;"
```

Take fresh backup

```
sudo /opt/backup/backup-db.sh
LATEST=$(ls -1t /backups/assessment_*.sql.gz | head -n1)
echo "Latest backup: $LATEST"
```

Create empty restore DB

```
docker exec -it postgres psql -U app_user -d assessment_db -c "DROP DATABASE IF EXISTS assessment_db_restore;"

docker exec -it postgres psql -U app_user -d assessment_db -c "CREATE DATABASE assessment_db_restore;"
```

Restore backup into restore DB

```
gunzip -c "$LATEST" | docker exec -i postgres psql -U app_user -d assessment_db_restore
```

Show row counts match (Counts must be the same)

```
docker exec -it postgres psql -U app_user -d assessment_db -c "SELECT count(*) FROM test_items;"

docker exec -it postgres psql -U app_user -d assessment_db_restore -c "SELECT count(*) FROM test_items;"
```

```
deploy@ip-172-31-39-3:~/devops-assessment$ docker exec -it postgres psql -U app_user -d assessment_db -c "SELECT 1;"
 ?column?
----------
        1
(1 row)
```

TASK 5 — Github action CICD

CI workflow (.github/workflows/ci.yml)

Triggers: push any branch + pull_request
Jobs: lint (ruff), test (pytest), docker-build
All must pass for PR merge

CD workflow (.github/workflows/deploy.yml)

Trigger: main only, after CI passes

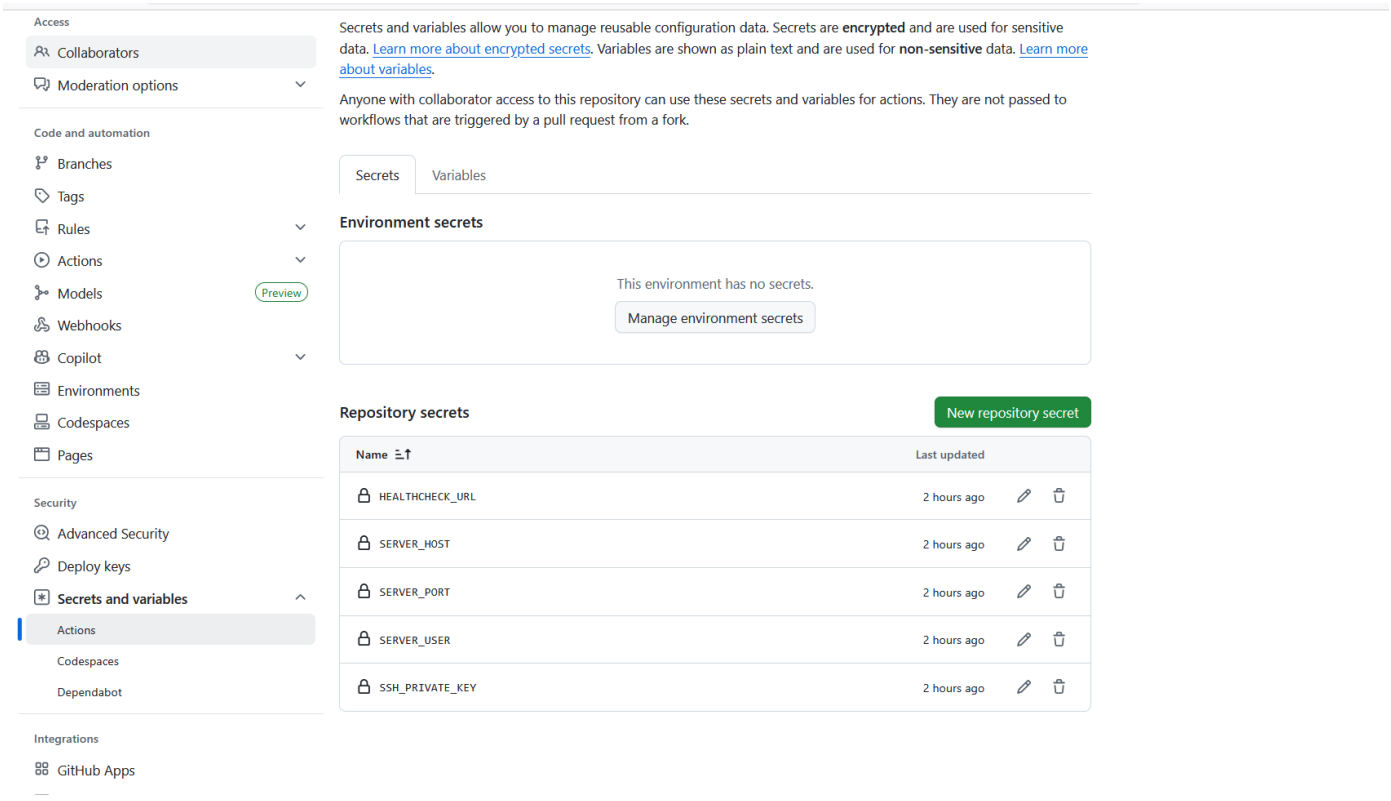
Steps:

- Build image tagged with git SHA
- SSH to EC2 using deploy key
- Pull new image + docker compose up -d --no-deps api
- Health check /health must return 200
- If fails: rollback to previous image tag

Secrets management

Store only in GitHub Secrets (no repo leaks):

- SERVER_HOST, SERVER_USER, SSH_PRIVATE_KEY, SERVER_PORT, HEALTHCHECK_URL

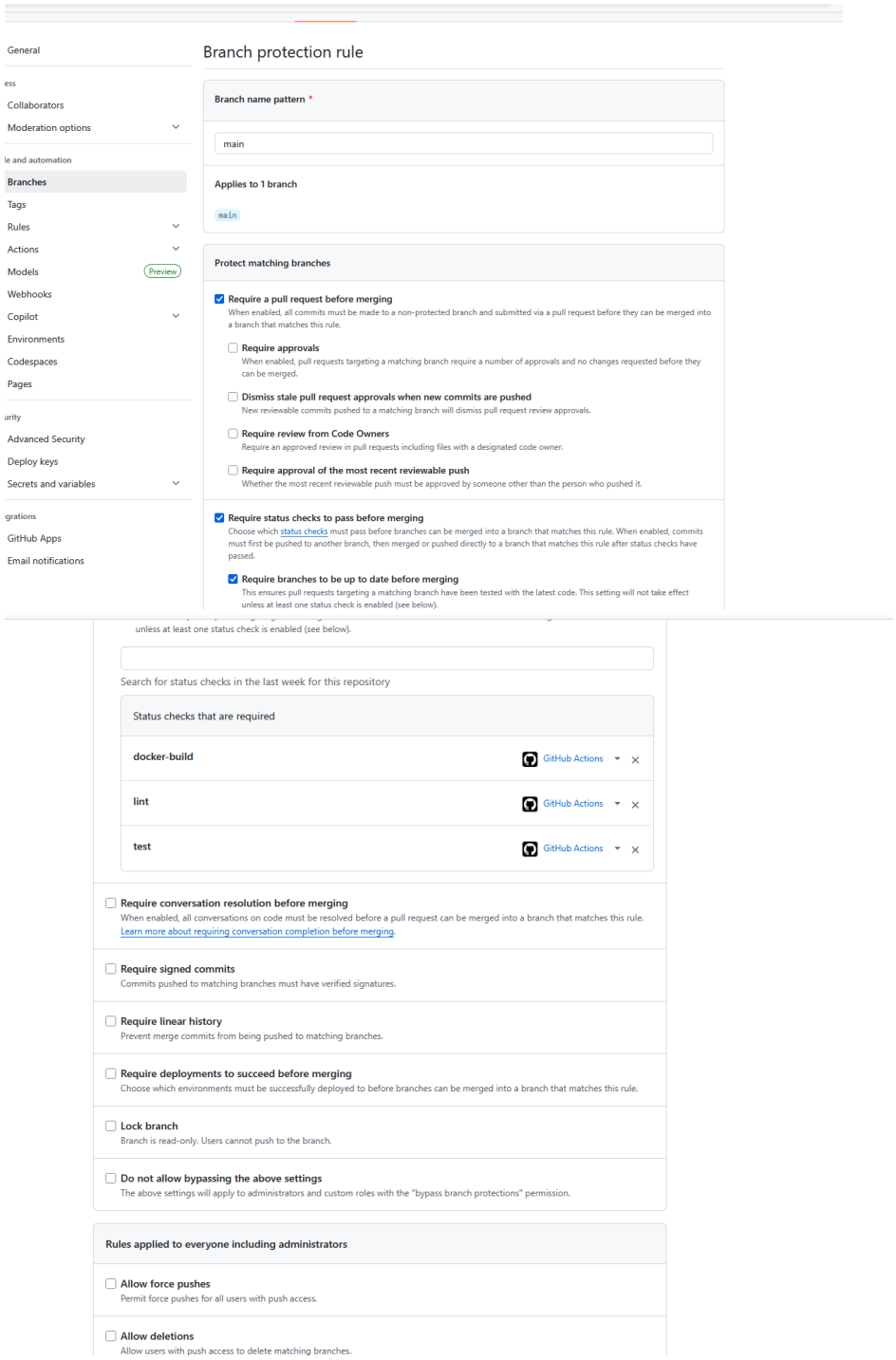


Branch protection

Enable on main:

- Require PR
- Require CI checks

- Prevent force pushes



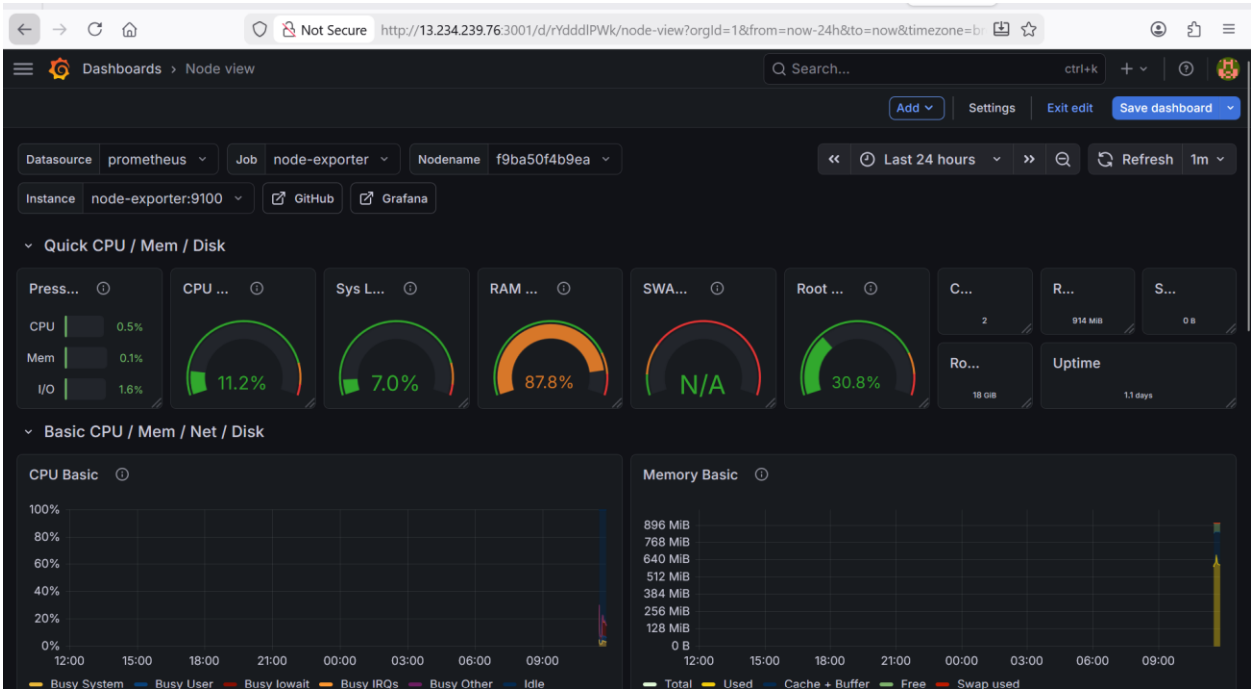
TASK 6 — Prometheus and Grafana

Prometheus + Node Exporter + cAdvisor

- Add services in docker-compose
- Prometheus scrapes node-exporter, cadvisor, api /metrics
- Prometheus NOT publicly exposed

Grafana

- Expose Grafana on non-standard port (3001)
- Add Prometheus datasource (http://prometheus:9090)
- Create/import dashboard: CPU, memory, disk, container status
- Export dashboard JSON and commit to repo



Alert rule (DiskSpaceHigh)

- prometheus/alerts.yml: disk usage > 75%

```
deploy@ip-172-31-39-3:~/devops-assessment/prometheus$ cat alerts.yml
groups:
  - name: host-alerts
    rules:
      - alert: DiskSpaceHigh
        expr: (1 - (node_filesystem_avail_bytes{fstype!~"tmpfs|overlay"} / node_filesystem_size_bytes{fstype!~"tmpfs|overlay"})) * 100 > 75
        for: 2m
        labels:
          severity: warning
        annotations:
          summary: "Disk space usage high (>75%) on {{ $labels.instance }}"
          description: "Disk usage is above 75% for more than 2 minutes."
```